RAHUL RAGHUDHAS

CLIMATE SCIENTIST | EARTH SYSTEM & OCEAN-ATMOSPHERE MODELING

PROJECT SCIENTIST-I
IITM, Pune | India
rahulraghudhas@gmail.com

PROFILE Climate scientist with research and model-development experience across IITM, NCMRWF and IIT Kharagpur. Expertise spans coupled Earth-system modeling, tropical climate dynamics, monsoon variability, decadal prediction and data-driven climate research.

CORE EXPERTISE

EARTH SYSTEM MODELING UFS, FV3/GFS v16, CICE6, MOM4p1	**TROPICAL CLIMATE** MJO, monsoon, tropical cyclones, PDO	**COUPLED MODELING** ROMS, WRF, COAWST
HPC & COUPLING MPI/OpenMP, ESMF, NUOPC, CMEPS/FMS	**CLIMATE DATA** Python, CDO, FERRET, GMT	**AI / STATISTICS** LSTM, CNN/BCNN, SHAP, ensemble analysis

PROFESSIONAL EXPERIENCE

Project Scientist-I | Indian Institute of Tropical Meteorology (IITM) | Pune
Jul 2025 - Present
Coupled climate-model development and prediction research involving CICE6/UFS, DCPP-C Pacific forcing, PDO-ISMR teleconnections.

Project Scientist-II | National Centre for Medium Range Weather Forecasting (NCMRWF) | Noida
Aug 2024 - Mar 2025
Worked under the Government of India Mission Monsoon-III project within NCMRWF's medium-range weather forecasting research programme.

Junior Project Assistant | Indian Institute of Technology Kharagpur | Kharagpur
Dec 2018 - Jul 2024
Research on MJO-tropical cyclone interactions and coupled air-sea processes using observations, climate diagnostics and COAWST/ROMS-WRF modeling.

Project Assistant | Indian Institute of Technology Kharagpur | Kharagpur
Jun 2017 - Dec 2017
Ocean-modeling research on shortwave penetration and upper-ocean response using ROMS.

M.Sc. Dissertation Researcher | Indian Institute of Technology Kharagpur | Kharagpur
Dec 2016 - May 2017
Numerical modeling of shortwave penetration in the Bay of Bengal using satellite optical datasets and ROMS.

EDUCATION

PhD Research Scholar — **Jul 2019 - Present**
CORAL, Indian Institute of Technology Kharagpur | MJO, tropical cyclones, Indo-Pacific warm pool and monsoon-cyclogenesis interactions

M.Sc. Oceanography — **2015 - 2017**
Cochin University of Science and Technology (CUSAT), Kerala | CGPA: 8.0/10

B.Sc. Physics — **2012 - 2015**
St. Dominic's College, Kanjirappally, Mahatma Gandhi University | 75.75%

SELECTED RESEARCH & MODEL DEVELOPMENT

DCPP-C NORTH PACIFIC / PDO-ISMR North Pacific forcing, PDO-like variability and teleconnection pathways to Indian summer monsoon rainfall.	**UFS / GFS v16 - IITM-ESM** UFS/S2SWAL and FV3/GFS v16 integration planning, component interfaces and coupled-model validation.
CICE6 SEA-ICE MODELING CICE6 configuration, climatological integrations, restart testing and coupled-model readiness on IITM HPC.	**MJO - TROPICAL CYCLONES** Basin-dependent MJO influence on tropical-cyclone genesis and activity across the Indo-Pacific.
INDO-PACIFIC WARM POOL Warm-pool expansion and associated changes in MJO variability and tropical-cyclone response.	**CLIMATE AI & PREDICTABILITY** Explainable LSTM for monsoon-cyclogenesis links and Bayesian CNN methods for probabilistic ENSO prediction.

TECHNICAL EXPERTISE

MODELING
CICE6, UFS, FV3/GFS v16, MOM4p1, ROMS, WRF, COAWST

COUPLING & HPC
ESMF, NUOPC, CMEPS/FMS, MPI/OpenMP, Linux HPC, PBS

PROGRAMMING & DATA
Python, Fortran, C/C++, MATLAB, Shell, CDO, FERRET, GMT

CLIMATE DIAGNOSTICS
MJO, ENSO, PDO, monsoon, tropical cyclones, sea ice, air-sea interaction

STATISTICS & AI
Regression, filtering, composites, significance tests, ensembles, LSTM, CNN/BCNN, SHAP

MODEL WORKFLOW
Compilation, dependencies, namelists, restart/reproducibility testing, output validation

MODEL DEVELOPMENT HIGHLIGHTS

CICE6 IMPLEMENTATION Completed long climatological integrations at 100 km and 25 km resolution with restart, reproducibility and diagnostic checks on IITM HPC.

UFS ON ARKA HPC compiled the coupled UFS framework and established the software/dependency environment for coupled model testing on IITM ARKA.

PEER-REVIEWED PUBLICATIONS

1. Rahul, R., Chakraborty, A., Kuttippurath, J., Almazroui, M., Alrasheedi, K. G., & Pillai, G. V. (2026). Decoding the Non-linear Impact of Monsoon on Cyclogenesis in the Bay of Bengal by Applying Deep Learning. Earth Systems and Environment. DOI: 10.1007/s41748-026-01196-2
2. Rahul, R., Kuttippurath, J., Chakraborty, A., & Akhila, R. S. (2022). The inverse influence of MJO on the cyclogenesis in the north Indian Ocean. Atmospheric Research, 265, 105880. DOI: 10.1016/j.atmosres.2021.105880

CURRENT RESEARCH OUTPUTS

MANUSCRIPT IN PREPARATION Impact of Aleutian Low-induced momentum and buoyancy forcing on Indian summer monsoon and associated teleconnection pathways.

PhD THESIS Influence of Madden-Julian Oscillation on the Tropical Cyclonic Activity over the Indian Ocean and the Pacific Ocean.

PROFESSIONAL & SCIENTIFIC DEVELOPMENT

INTROMET 2025 Participated in the meteorological research meeting at IITM Pune.

CURRENT FOCUS: DCPP-C Pacific forcing / PDO-ISMR teleconnection related studies.

LANGUAGES

Malayalam (native) | **English** (fluent) | **Hindi, Tamil** (basic)